When Lenders Disagree

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Abstract

I model banks' endogenous information acquisition on firms' types under asymmetric information and imperfect competition. Each bank independently chooses its screening precision and draws a binary signal on the borrower type that is independent of its rival's one, conditional on the borrower type. Disagreement implies that a bank that drew a good signal on a firm may be bidding against a rival that drew a bad one. Equilibrium rates are dispersed, and, when the signal is informative, banks post lower rates when observing a good signal. The value of information depends on whether the rate ranges associated with the two signals share a rate: in this case, an uninformed bank can earn the informed bank's profits by posting a shared rate. Therefore, with a positive and strictly increasing cost of information, only two market configurations survive: banks remain uninformed, or become informed when the two signals' rate ranges are disjoint.

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1. Introduction

Preview of results. I model banks' endogenous information acquisition about firms' types under asymmetric information and imperfect competition. Each bank independently chooses its screening precision q and draws a binary signal on the firm's type that, given the type, is independent of its rival's. The signal splits the firms a bank meets into a good-signal group and a bad-signal group. The two groups are not separate markets: the banks disagree on a share $2q(1-q)$ of firms, so a bank that drew a good signal on a firm may be bidding against a rival that drew a bad one, and the rates a bank posts to its bad-signal group compete with the rates its rival posts to its good-signal group.

Equilibrium rates are dispersed, and, when the signal is informative, banks post lower rates to their good-signal group. In each group, a bank's profit depends on two functions of the rate: what it earns on a good firm and what it earns on a bad firm. Their difference, the premium on a good firm, sorts the two groups: a bank posts the lowest rates in the market only to its good-signal group, and the highest rate always to its bad-signal group. How high the bad-signal group's rates go depends on its adverse-selection index, which compares what a bank earns from captive firms at the top of the pooling region, where it keeps the good firms, and at the top of the bad-only region, where it lends to bad firms only. When adverse selection in the bad-signal group is mild, a bank posts that group rates in the pooling region only. When it is severe, a share of those rates moves to the bad-only region, and all of them when it is severe enough. In the pooling region, the rate ranges associated with the two signals meet at one rate, overlap, or overlap after a pause, a band of rates posted to the bad-signal group alone. When all of the bad-signal group's rates are in the bad-only region, the two ranges are disjoint.

In a numerical example, as precision rises the rates posted to the good-signal group spread over the pooling region, while those posted to the bad-signal group are pushed to the top of the pooling region and then into the bad-only region, leaving a band of rates just above the pooling region that nobody posts. Profit in the good-signal group rises until the bad-signal group leaves the pooling region. Profit in the bad-signal group first falls, as fewer good firms are left in it, and recovers as its rates move to the bad-only region.

The value of information depends on whether the rate ranges associated with the two signals share a rate, as they do when they meet, overlap, or overlap after a pause. At a shared rate both groups earn their equilibrium profit. A bank that posts that rate to both its groups makes every firm the same offer, so its signal no longer matters and it still earns its equilibrium profit: an uninformed bank can earn the informed bank's profits. Screening less then costs nothing, although screening more can still pay. A bank loses profit by screening less only when the two ranges are disjoint.

Therefore, with a cost of precision that strictly increases, only two market configurations

survive: banks remain uninformed, or they are informed and the two signals' rate ranges are disjoint. In the second, a precision short of perfect survives only if the marginal cost of precision jumps at that precision, as with a screening technology that cannot go further. With a smooth cost no intermediate precision survives: banks remain uninformed, or their signal is perfect and the two ranges stay disjoint.

Related literature. I combine two explanations of price dispersion: the winner's curse among lenders with imperfect credit tests, and search frictions among buyers. I model information as in Broecker (1990): banks run imperfect and independent credit tests on each firm and compete in rates. A bank that cuts its rate wins firms to which its rival offered lower rates, and these are more often firms the rival judged good. Cutting the rate therefore improves the firms a bank lends to, which separates the problem from ordinary price competition. I combine this with models of price dispersion in which some buyers compare prices and others do not (Varian 1980; Burdett and Judd 1983; Narasimhan 1988). Neither literature alone gives the equilibrium of this note. Broecker's model has adverse selection but no search frictions: every borrower compares all offers, and rates are dispersed because of the winner's curse. The price-dispersion models have search frictions but no adverse selection: the cost of serving a buyer is the same whoever the buyer is. Hauswald and Marquez (2003, 2006) study endogenous screening when a bank's precision depends on its distance from the borrower. Lester, Shourideh, Venkateswaran and Zetlin-Jones (2019) also combine search frictions with adverse selection, but there the borrower holds the private information and lenders offer menus of contracts to induce self selection; here instead the lenders hold noisy signals.

The equilibrium within each signal group draws on a third set of papers, on price competition among sellers that differ (Baye, Kovenock and de Vries 1992; Golding and Slutsky 2000; and, in general form, Armstrong and Vickers 2022 and Myatt and Ronayne 2026). Within a signal group a bank prices like a seller with its own expected cost, so its two groups behave like a low-cost and a high-cost seller. These papers find that such sellers set prices on different intervals rather than on a common one, a pattern also seen in Bulow and Levin. The configurations here have the same shape: the good-signal group gets the low rates, the bad-signal group the high ones, and the two ranges meet, overlap or are disjoint. The argument that no rate in the pooling region goes unquoted is also the same. One feature does not carry over. In those papers a seller's cost does not depend on which buyers it wins. Here it does, because whether a bank wins a firm depends on the firm's type. A bank's profit in each group therefore depends on two functions of the rate, the value of a good firm and the value of a bad firm, not on one expected cost. Their difference, the premium on a good firm, decides which group gets which rates.

2. Environment

2.1 Banks, firms and signals

Banks and firms. Two risk-neutral banks meet a continuum of firms. Each firm borrows one unit at rate r. A share π of firms are shoppers: they get an offer from both banks and accept the lower one. The other $1-\pi$ are captive: they get one offer. A firm rejects any rate above its reservation rate.

Types. A firm is good ($\theta = g$, share μ_g) or bad ($\theta = b$, share $\mu_b = 1 - \mu_g$). Lending to it costs c_g or c_b; costs do not depend on the rate. Good firms accept rates up to $\bar{r}^g$ and bad firms up to $\bar{r}^b$, with

$$c_g < c_b < \bar{r}^g < \bar{r}^b.$$

Rates $r \leq \bar{r}^g$ form the **pooling region**, where both types accept. Rates $\bar{r}^g < r \leq \bar{r}^b$ form the **bad-only region**, where only bad firms accept.

Signals. Each bank runs its own risk assessment of the firm. I model it as a signal: bank $j \in \{1, 2\}$ draws its own signal $s_j \in \{g, b\}$ on the firm, with

$$\Pr(s_j = \theta \mid \theta) = q, \qquad q \in (\tfrac{1}{2}, 1].$$

Given the firm's type, s_1 and s_2 are independent. Precision is the same for both types and for both banks.

Signal groups. The signal splits the firms a bank meets into two groups: the **good-signal group**, the firms on which it drew g, and the **bad-signal group**, the firms on which it drew b. The shares of the two groups, and the share of good firms in each, are

$$\sigma_g = q\mu_g + (1-q)\mu_b, \quad \sigma_b = 1 - \sigma_g, \qquad \alpha^{s_g} = \frac{q\mu_g}{\sigma_g}, \quad \alpha^{s_b} = \frac{(1-q)\mu_g}{\sigma_b},$$

and the expected cost of lending to group s is $c^{avg,s} = \alpha^s c_g + (1-\alpha^s)c_b$. The signal sorts firms into groups but does not change how many good firms a bank meets in total:

$$\sigma_g \, \alpha^{s_g} + \sigma_b \, \alpha^{s_b} = \mu_g. \tag{1}$$

The two groups are not separate markets. Every firm sits in one group of each bank. Averaging over the type gives the joint distribution:

	rival's good-signal group	rival's bad-signal group
bank's good-signal group	$q^2\mu_g + (1-q)^2\mu_b$	$q(1-q)$
bank's bad-signal group	$q(1-q)$	$(1-q)^2\mu_g + q^2\mu_b$

The banks disagree on a share $2q(1-q)$ of firms, whatever μ_g: these firms sit in one bank's good-signal group and in the other's bad-signal group. So the rates a bank quotes its bad-signal group compete with the rates the rival quotes its good-signal group. For pricing, what matters is that given the firm's type, the rival's group does not depend on the bank's own: a good firm is in the rival's good-signal group with probability q, a bad firm with probability $1-q$.

Strategies and equilibrium. A bank draws the rate it quotes a firm in its signal group s from a distribution F^s with no atoms: F^{s_g} for the good-signal group, F^{s_b} for the bad-signal group. In a symmetric equilibrium both banks use the same pair. The profit $\Pi^s(r)$ on a firm of group s quoted r satisfies the equal-profit condition

$$\Pi^s(r) = \mathbf{\Pi}^s \text{ on the support of } F^s, \qquad \Pi^s(r) \leq \mathbf{\Pi}^s \text{ at every } r,$$

where $\mathbf{\Pi}^s$ is the equilibrium profit in group s. Rates in the market are distributed as $F = \sigma_g F^{s_g} + \sigma_b F^{s_b}$, and expected profit per firm examined is $\mathbb{E}[\Pi(q)] = \sigma_g \mathbf{\Pi}^{s_g} + \sigma_b \mathbf{\Pi}^{s_b}$.

Notation. A subscript g or b refers to the firm's type; a superscript s_g or s_b refers to the bank's signal group.

symbol	meaning
σ_s	share of signal group s
α^s, $c^{avg,s}$	share of good firms, and expected cost, in signal group s
F^s, $\mathbf{\Pi}^s$	rate distribution, and equilibrium profit, in signal group s
$Y_\theta(r)$	probability that the rival's offer to a type-θ firm is above r
$\Phi_\theta(r)$	profit on a type-θ firm quoted r
$\Delta(r)$	premium on a good firm, $\Phi_g(r) - \Phi_b(r)$
$A^s(r)$, $B^s(r)$	margin at stake in group s on firms in the rival's good- or bad-signal group
ϕ^s	adverse-selection index of signal group s
m	share of the bad-signal group's quotes in the bad-only region
$\underline{r}$, $\underline{r}^b$	lowest rate in the market; lowest rate quoted in the bad-only region
ρ, ρ_g, ρ_-, r_2	rates that decide the configuration (Section 4.3)

2.2 Profit in each signal group

The value of a good and of a bad firm. A shopper stays with a bank quoting r if the rival quotes more. The rival quotes from F^{s_g} or F^{s_b} depending on the group in which

it places the firm, and that depends on the firm's type. The probability that the rival's offer to a type-θ firm is above r is

$$Y_g(r) = q\left[1 - F^{s_g}(r)\right] + (1-q)\left[1 - F^{s_b}(r)\right],$$
$$Y_b(r) = (1-q)\left[1 - F^{s_g}(r)\right] + q\left[1 - F^{s_b}(r)\right],$$

and the bank's profit on a type-θ firm quoted r is

$$\Phi_\theta(r) = \left[1 - \pi + \pi\, Y_\theta(r)\right](r - c_\theta). \tag{2}$$

Lemma 1 (profit). In the pooling region a bank earns, on a firm of its signal group s,

$$\Pi^s(r) = \alpha^s\, \Phi_g(r) + (1-\alpha^s)\, \Phi_b(r), \qquad r \le \bar{r}^g.$$

In the bad-only region good firms reject, and $\Pi^s(r) = (1-\alpha^s)\, \Phi_b(r)$.

Both groups face the same two functions and differ only in the weights α^s. The lemma uses the independence of the signals given the type: once the type is known, the group in which one bank places a firm says nothing more about the group in which the rival places it.

The same profit, counted by the rival's group. A bank wins shoppers mostly when the firm sits in the rival's other group, and those firms are worse than the bank's own group suggests. The chance of winning and the margin therefore move together across types, so profit cannot be computed as an average chance of winning times an average margin. Counting firms by the rival's group rather than by their own type gives the same total; Appendix B.2 counts it both ways.

Compact form. Substituting Y_g and Y_b into Lemma 1 gives, in the pooling region,

$$\Pi^s(r) = (r - c^{avg,s}) - \pi\left[A^s(r)\, F^{s_g}(r) + B^s(r)\, F^{s_b}(r)\right], \tag{3}$$

$$A^s(r) = \alpha^s\, q\,(r - c_g) + (1-\alpha^s)(1-q)\,(r - c_b),$$
$$B^s(r) = \alpha^s(1-q)\,(r - c_g) + (1-\alpha^s)\, q\,(r - c_b).$$

The bank starts from the profit it would earn if every firm in the group accepted, $r - c^{avg,s}$. The rival's quotes to its good-signal group that lie below r take away the shoppers in that group; A^s is the margin at stake on those firms, and B^s the same for firms in the rival's bad-signal group. Profit is linear in the two distributions. Two numbers recur below:

$$L = 1 - \pi q, \qquad M = 1 - \pi + \pi q,$$

the chances of keeping a good firm and a bad firm when the rival quotes its whole good-signal group below r and its whole bad-signal group above.

3. Properties of equilibrium

Condition (A). In its good-signal group, a bank does not prefer to lend to bad firms only:

$$\mathbf{\Pi}^{s_g} \;\geq\; (1-\alpha^{s_g})(1-\pi)(\bar{r}^b - c_b).$$

Section 4.1 gives a sufficient condition that can be checked before solving. Everything below assumes (A). When (A) fails, both groups are quoted rates in the bad-only region; that case is not solved here (Section 4.4). Proof sketches are in Appendix A.

3.1 The premium sorts the two groups

Subtracting the two profits in Lemma 1,

$$\Pi^{s_g}(r) - \Pi^{s_b}(r) = (\alpha^{s_g} - \alpha^{s_b})\,\Delta(r), \qquad \Delta(r) \equiv \Phi_g(r) - \Phi_b(r). \tag{4}$$

Δ is the **premium on a good firm**, and it is the only thing on which the two groups differ. In its good-signal group a bank prefers rates at which the premium is high, in its bad-signal group rates at which it is low. Write $\Delta^* \equiv (\mathbf{\Pi}^{s_g} - \mathbf{\Pi}^{s_b})/(\alpha^{s_g} - \alpha^{s_b})$.

Lemma 2 (sorting). Where F^{s_g} has mass, $\Delta \geq \Delta^*$. Where F^{s_b} has mass, $\Delta \leq \Delta^*$. Where both have mass, $\Delta = \Delta^*$, and Φ_g and Φ_b are each constant:

$$\Phi_g = \frac{(1-\alpha^{s_b})\mathbf{\Pi}^{s_g} - (1-\alpha^{s_g})\mathbf{\Pi}^{s_b}}{\alpha^{s_g} - \alpha^{s_b}}, \qquad \Phi_b = \frac{\alpha^{s_g}\mathbf{\Pi}^{s_b} - \alpha^{s_b}\mathbf{\Pi}^{s_g}}{\alpha^{s_g} - \alpha^{s_b}}.$$

The last clause does the work: the two groups can be quoted the same range of rates only if neither the value of a good firm nor the value of a bad firm changes across it.

3.2 How the premium moves

Lemma 3. With $G_1(r) = q(r-c_g) - (1-q)(r-c_b)$ and $G_2(r) = (1-q)(r-c_g) - q(r-c_b)$,

$$\Delta(r) = (c_b - c_g) - \pi\left[F^{s_g}(r)\,G_1(r) + F^{s_b}(r)\,G_2(r)\right].$$

$G_1 > 0$, so quotes to the good-signal group placed below r always lower the premium at r. G_2 is positive below $r^\dagger = [qc_b - (1-q)c_g]/(2q-1)$ and negative above it, so quotes to the bad-signal group lower the premium at low rates and raise it at high rates. At the lowest rate in the market nobody has quoted yet, and the premium takes its largest value, $c_b - c_g$.

3.3 No rate in the pooling region goes unquoted

Lemma 4. In the pooling region, the union of the two groups' supports has no gap.

If nobody quoted on an interval (a, b) with quotes on both sides, a bank quoting a could move to just below b: it would face the same rival quotes and charge more.

3.4 Which group is quoted the lowest and the highest rates

Proposition 1. $\Delta^* < c_b - c_g$, and only the good-signal group is quoted the lowest rate in the market.

Proposition 2. The highest rate quoted to the good-signal group is no higher than the highest rate quoted to the bad-signal group.

3.5 The top of the support

At the highest rate in the market no rival quotes higher, so the bank earns only on captive firms.

Proposition 3 (the top of the support is a reservation rate). The highest rate in the market is quoted to the bad-signal group, and it is $\bar{r}^g$ or $\bar{r}^b$.

4. Construction of equilibrium

4.1 The adverse-selection index and the regimes of each group

Proposition 3 leaves two candidates for the top of the market. Comparing what the bad-signal group earns at each, from captive firms alone, gives its equilibrium profit:

$$\mathbf{\Pi}^{s_b} = \begin{cases} (1-\pi)(\bar{r}^g - c^{avg,s_b}) & \text{if } \phi^{s_b} \geq 0, \\ (1-\alpha^{s_b})(1-\pi)(\bar{r}^b - c_b) & \text{if } \phi^{s_b} < 0, \end{cases} \tag{5}$$

where, for signal group s,

$$\phi^s \equiv (\bar{r}^g - c^{avg,s}) - (1-\alpha^s)(\bar{r}^b - c_b) = \alpha^s(\bar{r}^g - c_g) - (1-\alpha^s)(\bar{r}^b - \bar{r}^g). \tag{6}$$

In the second form c_b drops out: a bad firm is served in both regions, so its cost cancels. Quoting $\bar{r}^g$ keeps the good firms of the group, with a margin $\bar{r}^g - c_g$ on each. Quoting $\bar{r}^b$ gives them up for the extra margin $\bar{r}^b - \bar{r}^g$ on bad firms. So $\phi^s \geq 0$ exactly when the odds that a firm in group s is good, $\alpha^s/(1-\alpha^s)$, are at least $(\bar{r}^b - \bar{r}^g)/(\bar{r}^g - c_g)$.

I call ϕ^s the **adverse-selection index** of signal group s. Adverse selection in the group is **mild** when $\phi^s > 0$: pooling at $\bar{r}^g$ earns more than lending to bad firms only at $\bar{r}^b$. It is **severe** when $\phi^s \leq 0$: lending to bad firms only earns at least as much.

The good-signal group pools. Condition (A) holds whenever $\phi^{s_g} \geq 0$, which covers every case of mild adverse selection in the good-signal group. The condition can be checked before solving: the group can always be quoted $\bar{r}^g$, where captive firms alone pay $(1-\pi)(\bar{r}^g - c^{avg,s_g})$.

Proposition 4 (the support is disconnected when a bank lends in both regions). Under (A), only the bad-signal group is quoted rates in the bad-only region. The share of its quotes there is

$$m = \min\left\{1,\ \max\left\{0,\ \frac{-(1-\pi)\,\phi^{s_b}}{\pi\, B^{s_b}(\bar{r}^g)}\right\}\right\}. \tag{7}$$

They lie on $[\underline{r}^b, \bar{r}^b]$, with

$$F^{s_b}(r) = 1 - \frac{1-\pi}{\pi q}\cdot\frac{\bar{r}^b - r}{r - c_b}, \qquad \underline{r}^b = \frac{(1-\pi)\,\bar{r}^b + \pi q\, m\, c_b}{1 - \pi + \pi q\, m},$$

and nobody quotes on $(\bar{r}^g, \underline{r}^b)$. So when $0 < m < 1$ the bad-signal group is quoted rates in both regions, and the support of F^{s_b} is disconnected.

In the bad-only region the rival quotes its whole good-signal group below, so a bank keeps a shopper from its bad-signal group only if the firm is also in the rival's bad-signal group (probability q for a bad firm) and the rival quotes higher. Profit in the group falls at $\bar{r}^g$, where the good firms leave, and climbs back to $\mathbf{\Pi}^{s_b}$ only at $\underline{r}^b$.

Three regimes of the bad-signal group. Equation (7) sorts the bad-signal group exactly as adverse selection sorts a market with no information. With mild adverse selection the group pools. With severe adverse selection a share m of its quotes moves to the bad-only region; m is zero at $\phi^{s_b} = 0$ and rises to one as ϕ^{s_b} falls:

regime	condition	the bad-signal group is quoted rates in
full pooling, $m = 0$	$\phi^{s_b} \geq 0$	the pooling region only
hybrid, $0 < m < 1$	$-\frac{\pi}{1-\pi}B^{s_b}(\bar{r}^g) < \phi^{s_b} < 0$	both regions
bad-only, $m = 1$	$\phi^{s_b} \leq -\frac{\pi}{1-\pi}B^{s_b}(\bar{r}^g)$	the bad-only region only

The threshold uses $B^{s_b}(\bar{r}^g)$, the margin on firms that sit in the bad-signal group of both banks. If the rival always placed the firm in the same group, as it does when the signal is public, $B^{s_b}(\bar{r}^g)$ would be $\bar{r}^g - c^{avg,s_b}$.

4.2 Rates quoted to both groups

By Lemma 2, Φ_g and Φ_b are constant on a range quoted to both groups. Their values are set at $\bar{r}^g$, where every quote to the good-signal group lies below and a share m of the quotes to the bad-signal group lies above:

$$\Phi_g^\top = \Big[1 - \pi + \pi(1-q)\,m\Big]\,(\bar{r}^g - c_g), \qquad \Phi_b^\top = \Big[1 - \pi + \pi q\,m\Big]\,(\bar{r}^g - c_b).$$

Proposition 5. On a range quoted to both groups,

$$\begin{aligned} F^{s_g}(r) &= \frac{1}{\pi}\Big[1 - \frac{1}{2q-1}\Big(\frac{q\,\Phi_g^\top}{r - c_g} - \frac{(1-q)\,\Phi_b^\top}{r - c_b}\Big)\Big], \\ F^{s_b}(r) &= \frac{1}{\pi}\Big[1 - \frac{1}{2q-1}\Big(\frac{q\,\Phi_b^\top}{r - c_b} - \frac{(1-q)\,\Phi_g^\top}{r - c_g}\Big)\Big], \end{aligned} \tag{8}$$

and $\mathbf{\Pi}^s = \alpha^s \Phi_g^\top + (1-\alpha^s)\Phi_b^\top$. Where the good-signal group is quoted alone, below every quote to the bad-signal group,

$$F^{s_g}(r) = \frac{(r - c^{avg,s_g}) - \mathbf{\Pi}^{s_g}}{\pi\, A^{s_g}(r)}, \qquad \underline{r} = c^{avg,s_g} + \mathbf{\Pi}^{s_g}. \tag{9}$$

When $m = 0$, each type of firm faces, on a shared range, the rate distribution of a one-type market with cost c_θ and top rate $\bar{r}^g$: $Y_\theta(r) = \frac{1-\pi}{\pi} \cdot \frac{\bar{r}^g - r}{r - c_\theta}$. The noise in the signals does not change the rates a type faces there; it changes in which group the firm is quoted them.

Proposition 6. A bank cannot withdraw quotes as the rate rises, so on a shared range both distributions must be non-decreasing. On (8), F^{s_b} always rises. F^{s_g} falls below

$$\rho_- = \frac{c_b - \tau c_g}{1 - \tau}, \qquad \tau = \sqrt{\frac{(1-q)\,\Phi_b^\top}{q\,\Phi_g^\top}},$$

rises above it, and equals one at $\bar{r}^g$. A shared range therefore ends at $\bar{r}^g$ and starts no lower than ρ_-.

4.3 The equilibrium

Let $\mathbf{\Pi}^\circ = \alpha^{s_g}\Phi_g^\top + (1-\alpha^{s_g})\Phi_b^\top$, the profit in the good-signal group at $\bar{r}^g$.

(iv) Bad-only, if $m = 1$. The bad-signal group is quoted only rates in the bad-only region. The good-signal group is quoted alone on the whole range $[\underline{r}, \bar{r}^g]$, from (9) with $\mathbf{\Pi}^{s_g} = \mathbf{\Pi}^\circ$, and F^{s_g} reaches one exactly at $\bar{r}^g$.

If $m < 1$, two rates decide the configuration.

- ρ is where quotes to the bad-signal group would start on the shared solution: $F^{s_b}(\rho) = 0$ in (8). It is the root in $(c_b, \bar{r}^g)$ of

$$q\left[(r - c_b) - \Phi_b^\top\right](r - c_g) = (1 - q)\left[(r - c_g) - \Phi_g^\top\right](r - c_b).$$

- ρ_g is where quotes to the good-signal group, alone at profit $\mathbf{\Pi}^\circ$, would run out: $F^{s_g}(\rho_g) = 1$ in (9), that is

$$\rho_g = \frac{\mathbf{\Pi}^\circ + \alpha^{s_g} L\, c_g + (1 - \alpha^{s_g}) M\, c_b}{\alpha^{s_g} L + (1 - \alpha^{s_g}) M}.$$

Proposition 7. With $m < 1$, the pooling region takes one of the following three configurations, and no other.

(i) Ranges meet, if $\rho_g < \rho$. Quotes to the good-signal group would run out before quotes to the bad-signal group start, leaving rates nobody quotes, which Lemma 4 rules out. Instead $\mathbf{\Pi}^{s_g}$ rises above $\mathbf{\Pi}^\circ$ until the good-signal group's range ends exactly where the bad-signal group's begins. At that rate $F^{s_g} = 1$ and $F^{s_b} = 0$, so

$$\rho = \frac{\mathbf{\Pi}^{s_b} + \alpha^{s_b} L\, c_g + (1 - \alpha^{s_b}) M\, c_b}{\alpha^{s_b} L + (1 - \alpha^{s_b}) M}, \qquad \mathbf{\Pi}^{s_g} = \alpha^{s_g} L\,(\rho - c_g) + (1 - \alpha^{s_g}) M\,(\rho - c_b).$$

The good-signal group is quoted on $[\underline{r}, \rho]$ from (9). The bad-signal group is quoted on $[\rho, \bar{r}^g]$ with

$$F^{s_b}(r) = \frac{\alpha^{s_b} L\,(r - c_g) + (1 - \alpha^{s_b}) M\,(r - c_b) - \mathbf{\Pi}^{s_b}}{\pi\, B^{s_b}(r)},$$

which reaches $1 - m$ at $\bar{r}^g$. The difference $\mathbf{\Pi}^{s_g} - \mathbf{\Pi}^\circ > 0$ is a rent: the good-signal group holds a band of rates on which the bad-signal group is not quoted.

(ii) Ranges overlap, if $\rho_g \geq \rho$ and $\rho \geq \rho_-$. $\mathbf{\Pi}^{s_g} = \mathbf{\Pi}^\circ$. The good-signal group is quoted alone on $[\underline{r}, \rho]$ from (9), and both groups are quoted on $[\rho, \bar{r}^g]$ from (8).

(iii) The good-signal group pauses, if $\rho_g \geq \rho$ and $\rho < \rho_-$. $\mathbf{\Pi}^{s_g} = \mathbf{\Pi}^\circ$. Sharing from ρ would require withdrawing quotes to the good-signal group. Instead F^{s_g} stays at

$\bar{F} = F^{s_g}(\rho)$ until (8) climbs back to that level, at $r_2 \in (\rho_-, \bar{r}^g)$. On the pause $[\rho, r_2]$ only the bad-signal group is quoted,

$$F^{s_b}(r) = \frac{(r - c^{avg,s_b}) - \mathbf{\Pi}^{s_b} - \pi \bar{F} A^{s_b}(r)}{\pi B^{s_b}(r)},$$

and both groups are quoted on $[r_2, \bar{r}^g]$ from (8). The pause is a band of rates never quoted to the good-signal group, although lower and higher rates are.

Regime and configuration. The regime of the bad-signal group (Section 4.1) does not fix the configuration. Configuration (iv) is the bad-only regime, but under full pooling and under hybrid lending each of the three configurations of the pooling region occurs. The table counts economies drawn at random that satisfy (A): μ_g and π uniform on $[0.05, 0.95]$, q uniform on $[0.51, 0.995]$, and $c_g < c_b < \bar{r}^g < \bar{r}^b$ the sorted values of four uniform draws on $[0, 1]$. In one economy of each cell, a brute-force check on a grid of rates confirms the equilibrium.

regime of the bad-signal group	(i) meet	(ii) overlap	(iii) pause	(iv) bad-only
full pooling	5,339	2,762	675	–
hybrid	2,071	2,631	453	–
bad-only	–	–	–	4,465

An example. I solve the equilibrium in the order set out in Appendix B.3. At every parameter value I report, I check the equal-profit condition by brute force: on a fine grid of rates, profit from Lemma 1 equals $\mathbf{\Pi}^s$ on each support and is no higher anywhere else (Appendix B.4 shows one case). The economy is $\mu_g = 0.65$, $c_g = 0.05$, $c_b = 0.30$, $\pi = 0.4$, $\bar{r}^g = 0.45$, $\bar{r}^b = 0.60$. Only the precision q varies. Condition (A) holds at every q in $[0.51, 0.99]$, and the brute-force check never finds an error above 10^{-15}. Figure 1 shows four equilibria, one in each configuration.

q	configuration	ϕ^{s_b}	m	$\mathbf{\Pi}^{s_g}$	$\mathbf{\Pi}^{s_b}$	$\mathbb{E}[\Pi(q)]$
0.70	(i) ranges meet	0.094	0	0.217	0.157	0.191
0.80	(iii) good-signal group pauses	0.024	0	0.222	0.138	0.188
0.90	(ii) ranges overlap	−0.056	0.706	0.244	0.149	0.208
0.97	(iv) bad-only	−0.120	1	0.243	0.170	0.217

q	rates quoted to the good-signal group	rates quoted to the bad-signal group
0.70	$[0.314, 0.395]$	$[0.395, 0.45]$
0.80	$[0.302, 0.398] \cup [0.425, 0.45]$	$[0.398, 0.45]$
0.90	$[0.308, 0.45]$	$[0.433, 0.45] \cup [0.511, 0.60]$
0.97	$[0.297, 0.45]$	$[0.482, 0.60]$

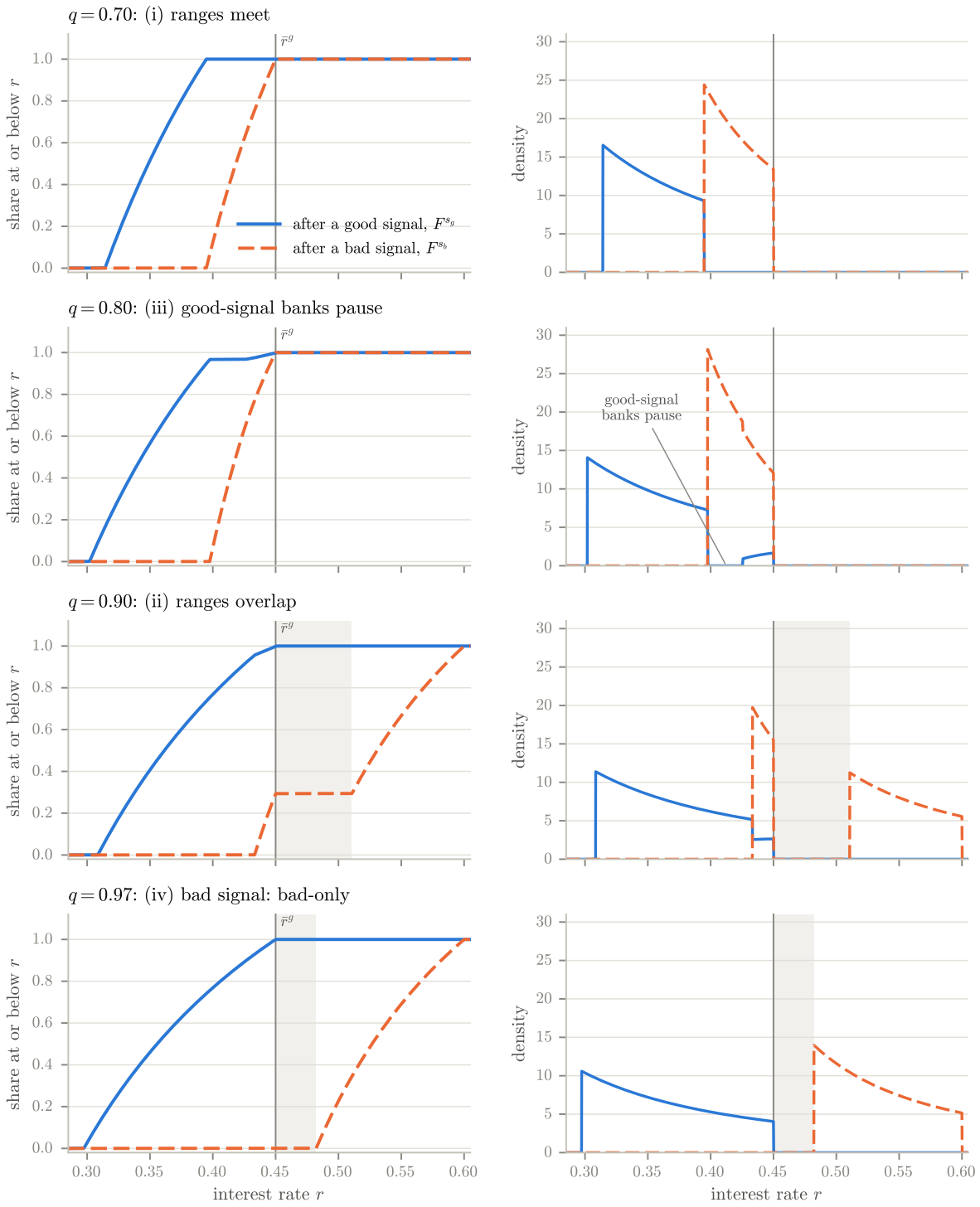


Figure 1: Equilibrium rate distributions at four values of q, one row each. Left: share of quotes at or below each rate. Right: density of quotes. Solid blue: good-signal group (F^{s_g}). Dashed orange: bad-signal group (F^{s_b}). Vertical line: $\bar{r}^g$, the top of the pooling region. Grey band: rates in the bad-only region that nobody quotes. Economy of Section 4.3.

- $q = 0.70$**, ranges meet.** The good-signal group is quoted rates on $[0.314, 0.395]$ and the bad-signal group on $[0.395, 0.45]$; the ranges touch at $\rho = 0.395$. A firm in the bad-signal group is still good with probability $\alpha^{s_b} = 0.44$, so adverse selection in the group is mild ($\phi^{s_b} > 0$) and it pools: nobody quotes in the bad-only region.
- $q = 0.80$**, the good-signal group pauses.** Quotes to the good-signal group run up to 0.398, stop, and resume on $[0.425, 0.45]$, which holds their last 3.3%. On the pause $[0.398, 0.425]$ only the bad-signal group is quoted; the density panel shows the gap in the blue curve.
- $q = 0.90$**, ranges overlap.** The good-signal group is quoted rates up to $\bar{r}^g$. The bad-signal group is hybrid: it shares $[0.433, 0.45]$ with the good-signal group and has 71% of its quotes in the bad-only region, on $[0.511, 0.60]$. Nobody quotes on $(0.45, 0.511)$.
- $q = 0.97$**, bad-only.** The good-signal group is quoted alone in the pooling region, on $[0.297, 0.45]$. The bad-signal group is quoted only rates that bad firms alone accept, on $[0.482, 0.60]$.

4.4 Limits: no information and a perfect signal

No information, $q = \frac{1}{2}$**.** Both groups carry the belief μ_g and $Y_g = Y_b$. Lemma 1 becomes the profit of a market with no information: $[1 - \pi + \pi(1 - F(r))](r - c^{avg})$ in the pooling region and $\mu_b[1 - \pi + \pi(1 - F(r))](r - c_b)$ in the bad-only region. One distribution serves both groups, so the equilibrium is the no-information one. As q falls to $\frac{1}{2}$ the equilibrium of this section converges to it: in the economy of Section 4.3 (full pooling at $q = \frac{1}{2}$), I find $\mathbf{\Pi}^{s_g} = 0.18752$ and $\mathbf{\Pi}^{s_b} = 0.18749$ at $q = 0.5001$, against $(1 - \pi)(\bar{r}^g - c^{avg}) = 0.1875$ with no information. Near $q = \frac{1}{2}$, condition (A) depends on the regime that prevails at $q = \frac{1}{2}$. At $q = 0.52$ it holds in every random economy I draw with full pooling, in most with hybrid lending, and in none with bad-only lending. In those economies I do not solve the equilibrium.

Perfect signal, $q = 1$**.** $\alpha^{s_g} = 1$, $\alpha^{s_b} = 0$, $Y_g = 1 - F^{s_g}$ and $Y_b = 1 - F^{s_b}$. Each group competes only with the rival's group of the same signal, and the model splits into two one-type markets, one per type. As $q \to 1$ the equilibrium converges to that of a perfect signal. The good-signal group earns $\mathbf{\Pi}^{s_g} \to (1 - \pi)(\bar{r}^g - c_g)$ and is quoted the good-type distribution of that equilibrium. In the bad-signal group adverse selection becomes severe, $\phi^{s_b} \to \bar{r}^g - \bar{r}^b < 0$, and $B^{s_b}(\bar{r}^g) \to \bar{r}^g - c_b$, so

$$m \to \min\left\{1,\ \frac{(1 - \pi)(\bar{r}^b - \bar{r}^g)}{\pi\,(\bar{r}^g - c_b)}\right\}, \qquad \underline{r}^b \to \max\left\{\bar{r}^g,\ c_b + (1 - \pi)(\bar{r}^b - c_b)\right\}.$$

The bad-signal group is quoted the bad-type distribution of a perfect signal, on $[c_b + (1 -$

$\pi)(\bar{r}^b - c_b), \bar{r}^b]$. When the lowest of these rates lies below $\bar{r}^g$, $m < 1$ in the limit: with no good firms left in the group, a bank loses nothing at $\bar{r}^g$, and the band nobody quotes closes. So $m = 1$ in the limit only if $(1-\pi)(\bar{r}^b - \bar{r}^g) \geq \pi(\bar{r}^g - c_b)$.

5. The equilibrium as precision rises

Figure 2 follows the equilibrium of the economy of Section 4.3 as q rises from 0.51 to 0.99.

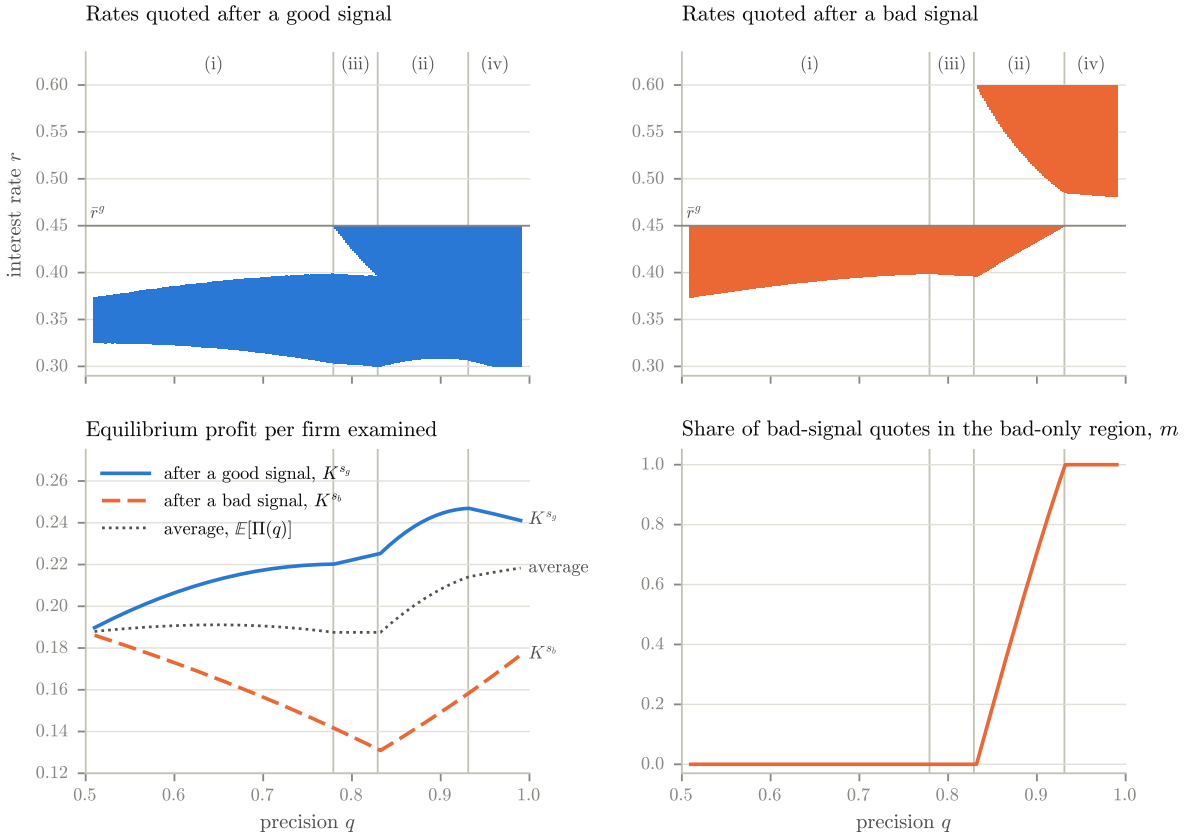


Figure 2: The equilibrium as q rises from 0.51 to 0.99, other parameters fixed. Top: rates quoted to the good-signal group (left) and to the bad-signal group (right), with the configuration above each range of q. Bottom left: equilibrium profit in each signal group and expected profit. Bottom right: share of the bad-signal group's quotes in the bad-only region. Vertical lines: changes of configuration, at $q = 0.78$, 0.83 and 0.93. Horizontal line: $\bar{r}^g$. Economy of Section 4.3.

- $q < 0.78$**: ranges meet.** The good-signal group is quoted the low rates and the bad-signal group the rates up to $\bar{r}^g$. The meeting rate rises from 0.373 to 0.399.
- $0.78 \leq q < 0.83$**: the good-signal group pauses.** The pause opens just below $\bar{r}^g$ ($[0.399, 0.449]$ at $q = 0.78$) and shrinks as q rises ($[0.397, 0.405]$ at $q = 0.82$) until it closes.
- $0.83 \leq q < 0.93$**: ranges overlap.** The good-signal group is quoted rates up to $\bar{r}^g$. From $q = 0.834$, adverse selection in the bad-signal group is severe, $\phi^{s_b} < 0$: the

group turns from full pooling to hybrid, m rises from 0 to 1, and its range in the pooling region shrinks toward $\bar{r}^g$.

- $q \geq 0.93$**: bad-only.** The bad-signal group is quoted rates on roughly $[0.48, 0.60]$.

Profits:

- $\mathbf{\Pi}^{s_g}$ rises from 0.190 at $q = 0.51$ to 0.247 at $q = 0.93$, then falls to 0.241.
- $\mathbf{\Pi}^{s_b}$ falls from 0.186 to 0.131 at $q = 0.83$, while the bad-signal group still pools with fewer and fewer good firms in it. It then rises to 0.177 as the group's quotes move to the bad-only region.
- Expected profit per firm examined stays between 0.187 and 0.191 up to $q = 0.83$, then rises to 0.218 at $q = 0.99$.

6. Information acquisition: profitable deviations

I ask whether a market's precision q^* is an equilibrium of the information-acquisition game: can a bank gain by choosing its own precision q_1 while its rival keeps q^* and the equilibrium rates $F^s(\cdot\,; q^*)$? With private signals the question has an exact answer.

6.1 The deviating bank's problem

What stays fixed. The rival keeps precision q^* and the rates of the equilibrium at q^*. The two banks' signals are still independent given the firm's type. Holding the rival's rates fixed is the right test when the rival cannot see q_1 before it sets its rates.

What does not change. Given the firm's type, the group in which the rival places it depends only on q^*. So Y_g, Y_b and the values Φ_g, Φ_b are those of the equilibrium at q^*, with $\Phi_g = 0$ in the bad-only region.

What changes. Only the deviator's own groups: their shares and the share of good firms in each, $\sigma_g(q_1)$, $\alpha^{s_g}(q_1)$ and $\alpha^{s_b}(q_1)$, from Section 2.1 with q_1 in place of q. In a group with share α of good firms, the deviator quotes its best rate and earns

$$W(\alpha) = \max_r \ \Big[\alpha\, \Phi_g(r) + (1-\alpha)\, \Phi_b(r)\Big].$$

Its expected profit per firm examined is

$$\Pi^{dev}(q_1) = \sigma_g(q_1)\, W\Big(\alpha^{s_g}(q_1)\Big) + \sigma_b(q_1)\, W\Big(\alpha^{s_b}(q_1)\Big), \tag{10}$$

with $\Pi^{dev}(q^*) = \mathbb{E}[\Pi(q^*)]$. The case $q_1 = \frac{1}{2}$ is a bank that does not consult a signal: its two groups have the same share μ_g of good firms. The participation decision, whether to pay C_s for the signal, compares $\Pi^{dev}(q^*) - C_s$ with $\Pi^{dev}(\frac{1}{2})$.

When the two precisions differ, a firm on which the banks disagree is no longer good with probability μ_g: the signal of the more precise bank counts more. At $q^* = 0.9$, a firm in the deviator's good-signal group and in the rival's bad-signal group is good with probability 0.95 if $q_1 = 0.99$ and 0.45 if $q_1 = 0.80$. A bank wins shoppers mostly when the rival disagrees, so a sharper signal improves the firms it wins.

6.2 How deviation profit depends on precision

Let D^+ and D^- denote the slopes of Π^{dev} just above and just below q^*.

Proposition 8. Π^{dev} is convex and non-decreasing in q_1, and

$$D^{\pm} = \mu_g\Big[\Phi_g(r_g^{\pm}) - \Phi_g(r_b^{\pm})\Big] - \mu_b\Big[\Phi_b(r_g^{\pm}) - \Phi_b(r_b^{\pm})\Big]. \tag{11}$$

A little more precision raises the share of good firms in the good-signal group and lowers it in the bad-signal group. By (4), the deviator then quotes its good-signal group the rate with the highest premium in that group's equilibrium range, $r_g^+ = \underline{r}$, and its bad-signal group the rate with the lowest premium in that group's range, r_b^+. A little less precision does the opposite: r_g^- has the lowest premium in the good-signal group's range and r_b^- the highest in the bad-signal group's range. Two closed forms follow:

- If $m > 0$, every rate in the bad-only range has the lowest premium, with $\Phi_g = 0$ and $\Phi_b = (1-\pi)(\bar{r}^b - c_b)$:

$$D^+ = \mu_g\,(\underline{r} - c_g) - \mu_b\Big[(\underline{r} - c_b) - (1-\pi)(\bar{r}^b - c_b)\Big].$$

- In configuration (iv), $r_g^- = \bar{r}^g$ and r_b^- is any rate in the bad-only range:

$$D^- = \mu_g\,(1-\pi q^*)(\bar{r}^g - c_g) - \mu_b\Big[(1-\pi+\pi q^*)(\bar{r}^g - c_b) - (1-\pi)(\bar{r}^b - c_b)\Big].$$

If $m = 0$, the lowest premium in the bad-signal group's range need not be at the top of the market. In configuration (iii) it lies inside the pause.

Proposition 9 (less precision). $\Pi^{dev}(q_1) = \Pi^{dev}(q^*)$ for every $q_1 \le q^*$ if and only if some rate $r^\sharp$ is quoted to both groups in equilibrium, which is the case exactly when $m < 1$, in configurations (i) to (iii). The deviator then quotes $r^\sharp$ to both its groups and

earns $\mu_g \Phi_g(r^\sharp) + \mu_b \Phi_b(r^\sharp)$, which by (1) does not depend on q_1. In configuration (iv), $\Pi^{dev}(q_1) < \Pi^{dev}(q^*)$ for every $q_1 < q^*$, although Π^{dev} can be flat over a range of low q_1.

At a rate quoted to both groups, profit depends only on how many good firms the bank lends to, which is μ_g whatever its precision. Competition has already equalized the two groups' profits there, so the information is worth nothing. For the participation decision: whenever the two groups share a rate, a bank that stops paying for the signal earns what a bank that pays earns. No bank then pays any $C_s > 0$, and a market in which both banks consult the signal unravels.

6.3 Which markets sustain their precision

Let $C(q)$ be the cost of precision q, strictly increasing. The market's precision q^* survives if no bank gains by deviating once the cost is counted:

$$\Pi^{dev}(q_1) - C(q_1) \;\le\; \Pi^{dev}(q^*) - C(q^*) \qquad \text{for every } q_1.$$

Proposition 10. For $q^* > \frac{1}{2}$, a strictly increasing convex cost that sustains q^* exists if and only if $\Pi^{dev}(q_1) < \Pi^{dev}(q^*)$ for every $q_1 < q^*$.

The construction is a cost that rises by S_{down} per unit of precision below q^* and by S_{up} above it, with

$$S_{down} = \frac{\Pi^{dev}(q^*) - \Pi^{dev}(\frac{1}{2})}{q^* - \frac{1}{2}}, \qquad S_{up} = \frac{\Pi^{dev}(1) - \Pi^{dev}(q^*)}{1 - q^*}.$$

Because Π^{dev} is convex, S_{down} is the smallest loss per unit of precision dropped and S_{up} the largest gain per unit added, so $S_{up} \ge S_{down}$.

Proposition 11. Only two kinds of market sustain their precision against some strictly increasing convex cost: $q^* = \frac{1}{2}$, and configuration (iv), that is $\phi^{s_b} \le -\frac{\pi}{1-\pi} B^{s_b}(\bar{r}^g)$. Markets in which the ranges meet, overlap, or are separated by a pause are ruled out: a bank lowers its precision, quotes the shared rate to both its groups, loses nothing, and saves cost.

In configuration (iv) no smooth cost works either: Π^{dev} has a corner at q^* with $D^- < D^+$, so a differentiable cost would need a marginal cost at q^* no larger than D^- and no smaller than D^+. The cost must rise slowly up to q^* and steeply beyond it, as with a screening technology that cannot go past q^*.

6.4 The two ends

Proposition 12 (starting from no information). At $q^* = \frac{1}{2}$, with $\mu_b = 1 - \mu_g$,

$$D^+(\tfrac{1}{2}) = 2\mu_g\mu_b\left[W'_+(\mu_g) - W'_-(\mu_g)\right] = \begin{cases} 2\pi\,\mu_g\mu_b\,(c_b - c_g) & \text{full pooling,} \\ 2\mu_g\mu_b\left[(c_b - c_g) + (1-\pi)(\bar{r}^b - c_b)\right] & \text{hybrid,} \\ 0 & \text{bad-only,} \end{cases}$$

where the regime is the one that prevails at $q = \frac{1}{2}$. The market with no information survives if and only if $C'(\frac{1}{2}) \geq D^+(\frac{1}{2})$. Under full pooling the gain from the first unit of precision is the product of three things: the share of shoppers (without them rates are not dispersed), $\mu_g\mu_b$ (there is nothing to learn about a pool of one type), and $c_b - c_g$ (what is at stake per firm). Under bad-only lending no good firm borrows, so there is nothing to sort and the first unit of precision is worth nothing.

Perfect signal. At $q^* = 1$, if $(1-\pi)(\bar{r}^b - \bar{r}^g) \geq \pi(\bar{r}^g - c_b)$, the bad-signal group is quoted only rates in the bad-only region (Section 4.4) and

$$D^-(1) = \mu_g\,(1-\pi)(\bar{r}^g - c_g) - \mu_b\Big[(\bar{r}^g - c_b) - (1-\pi)(\bar{r}^b - c_b)\Big].$$

A perfect signal survives if $C'(1^-) \leq D^-(1)$ (and the whole saving from any lower precision stays below the profit lost). If instead $(1-\pi)(\bar{r}^b - \bar{r}^g) < \pi(\bar{r}^g - c_b)$, the good-type and bad-type ranges share rates, Π^{dev} is flat on $[\frac{1}{2}, 1]$, and no increasing cost sustains a perfect signal.

In the economy of Section 4.3, $D^+(\frac{1}{2}) = 0.0455$ (full pooling) and $D^-(1) = 0.167$; a bank that drops from $q^* = 1$ to no information loses 4.8% of profit.

6.5 Simulation: deviations from four equilibria

The rival sits at each of the four equilibria of Section 4.3. Gains and losses are in % of $\Pi^{dev}(q^*)$, before the cost of precision.

q^*	configuration	$\Pi^{dev}(q^*)$	lower to $q_1 = 0.5$	raise to $q_1 = 0.999$	D^-	D^+
0.70	(i) ranges meet	0.191	0%	+23.1%	0	0.042
0.80	(iii) pause	0.188	0%	+20.8%	0	0.039
0.90	(ii) ranges overlap	0.208	0%	+10.8%	0	0.228
0.97	(iv) bad-only	0.217	−2.8%	+3.0%	0.170	0.225

- **Lowering precision.** In configurations (i) to (iii) profit does not change. The deviator quotes a shared rate to both its groups: 0.395 at $q^* = 0.70$, 0.398 at $q^* = 0.80$, and 0.433 at $q^* = 0.90$. At $q^* = 0.97$ no rate is shared. For q_1 below 0.935 the deviator quotes $\bar{r}^g$ to both groups and loses 2.8%; between 0.935 and 0.97

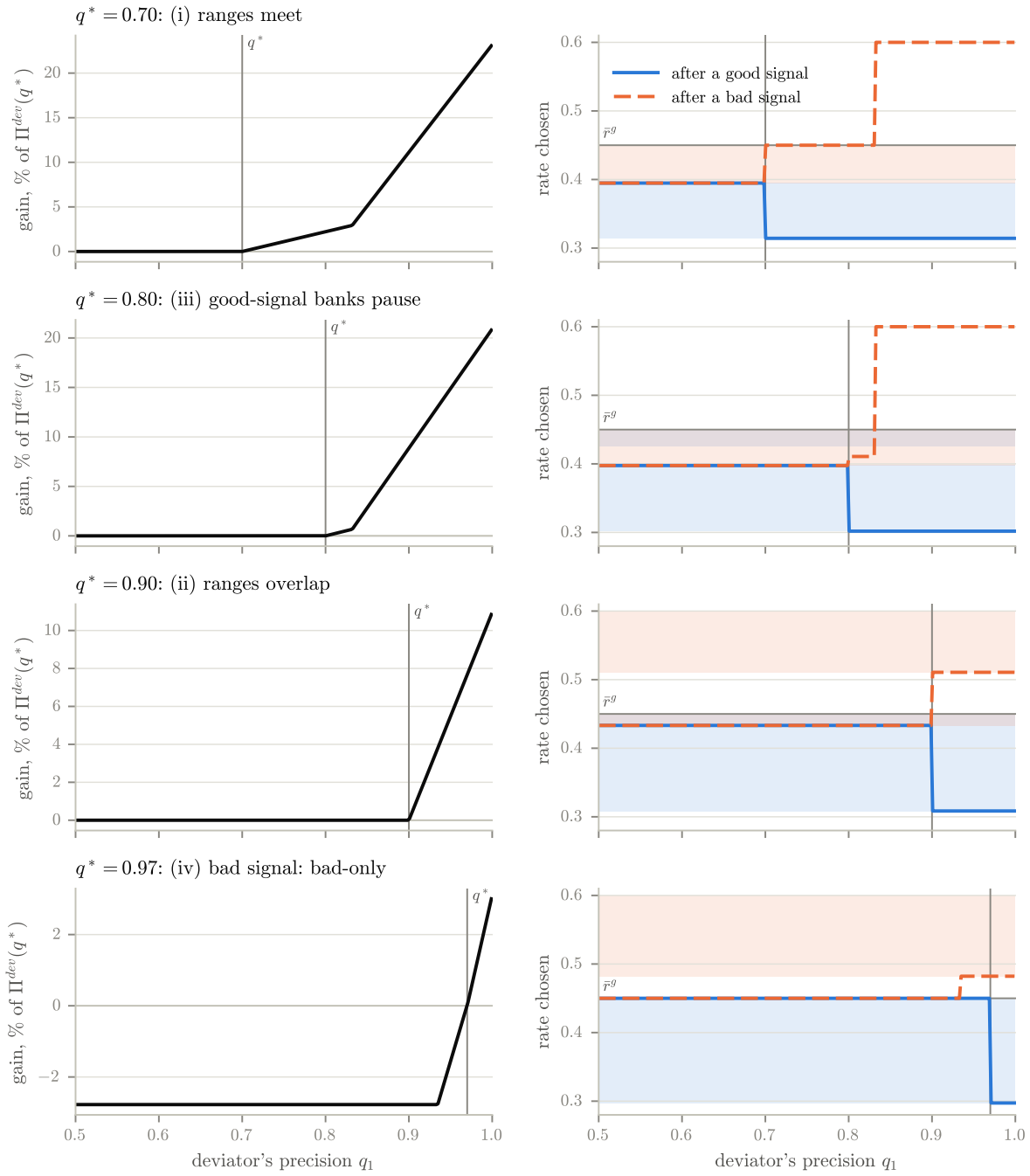


Figure 3: One bank deviates to precision q_1 while the rival stays at the equilibrium with precision q^* (vertical line), for the four equilibria of Section 4.3. Left: change in deviation profit, $\Pi^{dev}(q_1) - \Pi^{dev}(q^*)$, in % of $\Pi^{dev}(q^*)$, before the cost of precision. Right: rate the deviator quotes its good-signal group (solid blue) and its bad-signal group (dashed orange); when several rates tie, the lowest. Light bands: rates quoted in equilibrium to the good-signal group (blue) and to the bad-signal group (orange); where the bands overlap, both groups are quoted. Horizontal line: $\bar{r}^g$. Economy of Section 4.3.

it quotes $\bar{r}^g$ to its good-signal group and 0.482 to its bad-signal group, and the loss shrinks to zero.

- **Raising precision.** The deviator quotes its good-signal group the lowest rate in the market: 0.314, 0.302, 0.308 and 0.297. Its bad-signal group is quoted $\bar{r}^g$ at $q^* = 0.70$, and 0.411, inside the pause (Proposition 8), at $q^* = 0.80$. In both markets the deviator moves that group to $\bar{r}^b = 0.60$ from $q_1 = 0.83$ on, lending to its bad firms only, which is the kink in the gain. At $q^* = 0.90$ and 0.97 the bad-signal group is quoted in the bad-only range (0.511 and 0.482). The slopes in the table match (11).

6.6 Simulation: deviations at every market precision

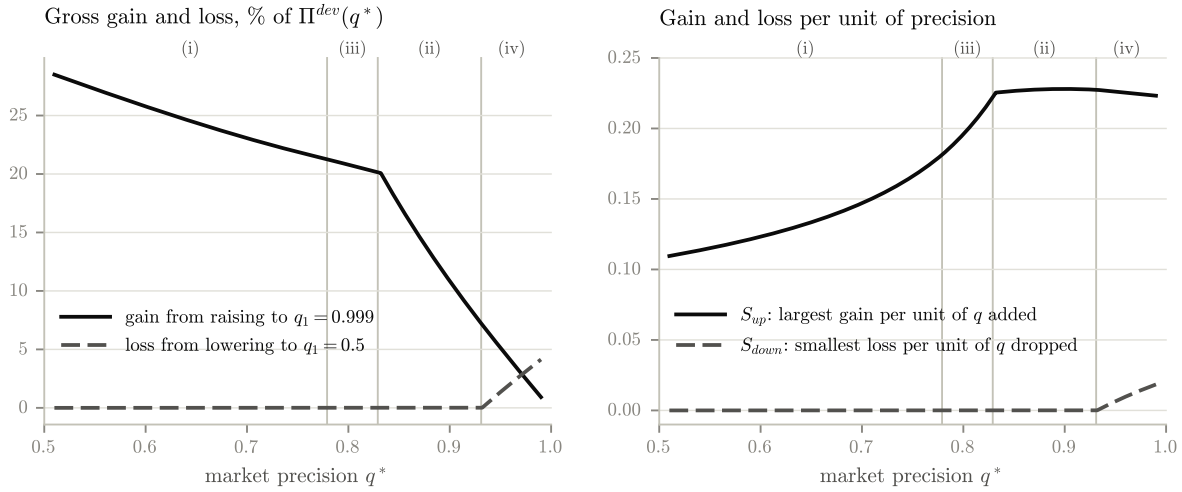


Figure 4: Gains and losses from a one-bank deviation, at every equilibrium precision q^* of the economy of Section 4.3. Left: gain from raising precision to $q_1 = 0.999$ and loss from lowering it to $q_1 = 0.5$, in % of $\Pi^{dev}(q^*)$, before the cost of precision. Right: S_{up} and S_{down}, with $q_1 = 0.999$ in place of 1. Vertical lines and labels: configurations, as in Figure 2.

- The gain from raising precision to 0.999 falls from 28.5% of profit at $q^* = 0.51$ to 7.3% at $q^* = 0.93$ and 0.9% at $q^* = 0.99$.
- The loss from lowering precision to 0.5 is zero up to $q^* = 0.930$ (configurations (i) to (iii)). It then rises to 4.1% at $q^* = 0.99$, and exceeds the gain from raising from $q^* = 0.972$ on.
- S_{up} stays between 0.11 and 0.23 at every q^*. S_{down} is zero up to $q^* = 0.930$ and stays below 0.02.
- In configuration (iv), $D^+ - D^-$ is at least 0.05 at every $q^* \geq 0.932$: a cost that sustains these markets must have a marginal cost that jumps at q^* by at least that much.

Appendix A. Proof sketches

Lemma 1. Profit is the expectation over the firm's type of the chance of keeping the firm times the margin. Given the type, the rival places the firm in its good-signal group with probability q for a good firm and $1-q$ for a bad firm, whatever the bank's own group; averaging the rival's distribution over its two groups gives Y_θ. A captive firm accepts and a shopper accepts when not undercut, so the chance of keeping a type-θ firm is $1-\pi+\pi Y_\theta$ and the margin is $r-c_\theta$. The bank's own group enters only through the weights α^s. In the bad-only region good firms reject, which deletes the first term.

Lemma 2. Equation (4) is Lemma 1 differenced. Where F^{s_g} has mass, $\Pi^{s_g} = \mathbf{\Pi}^{s_g}$ and $\Pi^{s_b} \leq \mathbf{\Pi}^{s_b}$, so $(\alpha^{s_g} - \alpha^{s_b})\Delta \geq \mathbf{\Pi}^{s_g} - \mathbf{\Pi}^{s_b}$. The same argument gives the second claim, and the two together give the third. Where both have mass, the two conditions $\alpha^s\Phi_g + (1-\alpha^s)\Phi_b = \mathbf{\Pi}^s$ are linear in (Φ_g, Φ_b) with determinant $\alpha^{s_g} - \alpha^{s_b} \neq 0$, so their solution is unique and does not depend on r.

Lemma 3. Substitute Y_g and Y_b into $\Delta = \Phi_g - \Phi_b$ and collect the terms in F^{s_g} and F^{s_b}. $G_1 > 0$ because $q > 1-q$ and $r - c_g > r - c_b$. G_2 is linear with slope $-(2q-1)$ and is zero at $r^\dagger$.

Lemma 4. Suppose nobody quotes on (a, b) in the pooling region while a is in some support. Moving those quotes from a to just below b leaves Y_g and Y_b unchanged. With Y_g, Y_b fixed, the derivative of Π^s in r is $\alpha^s[1-\pi+\pi Y_g] + (1-\alpha^s)[1-\pi+\pi Y_b] > 0$, so the move raises profit above $\mathbf{\Pi}^s$, which contradicts equilibrium.

Proposition 1. Let R_g be the top of the good-signal group's range, where $F^{s_g} = 1$. By Lemma 3, $\Delta(R_g) = (c_b - c_g) - \pi[G_1 + F^{s_b}G_2]$, and $G_1 + F^{s_b}G_2 \geq \min\{G_1, G_1 + G_2\} > 0$ because $G_1 + G_2 = c_b - c_g$. So $\Delta(R_g) < c_b - c_g$, and Lemma 2 gives $\Delta^* \leq \Delta(R_g) < c_b - c_g$. At the lowest rate both distributions are zero, so $\Delta = c_b - c_g > \Delta^*$: by Lemma 2 the bad-signal group cannot be quoted there.

Proposition 2. Suppose the top of the good-signal group's range, R_g, exceeds the top of the bad-signal group's range, R_b. At R_g nobody quotes higher, so $\Delta(R_g) = (1-\pi)(c_b - c_g)$. At R_b, $F^{s_b} = 1$ and $F^{s_g} = 1-u$ with $u > 0$, so $\Delta(R_b) = (1-\pi)(c_b - c_g) + \pi u\, G_1(R_b) > \Delta(R_g)$. Both groups are quoted near R_b, so $\Delta(R_b) = \Delta^*$ and $\Delta(R_g) < \Delta^*$. But only the good-signal group is quoted on $(R_b, R_g]$, where Lemma 2 requires $\Delta \geq \Delta^*$: a contradiction.

Proposition 3 and equation (5). Let R be the highest rate in the market; by Proposition 2 it is quoted to the bad-signal group, and nobody quotes above it. If $R < \bar{r}^g$, quoting slightly above R earns $(1-\pi)(R + \epsilon - c^{avg,s_b}) > \mathbf{\Pi}^{s_b}$; the same holds inside the bad-only region. So $R \in \{\bar{r}^g, \bar{r}^b\}$, and Lemma 1 with $Y_g = Y_b = 0$ gives (5). $R = \bar{r}^g$ requires that quoting $\bar{r}^b$ earn no more, $(1-\alpha^{s_b})(1-\pi)(\bar{r}^b - c_b) \leq (1-\pi)(\bar{r}^g - c^{avg,s_b})$,

which is $\phi^{s_b} \geq 0$. Expanding $c^{avg,s}$ gives the second form of (6).

Proposition 4. In the bad-only region the value of a bad firm is at most $(1-\pi)(\bar{r}^b - c_b)$: that is its value on the bad-signal group's range there, and it is lower on the band nobody quotes. In its good-signal group a bank therefore earns at most $(1-\alpha^{s_g})(1-\pi)(\bar{r}^b - c_b)$ there, which (A) rules out. With every quote to the good-signal group below, $Y_b = q(1-F^{s_b})$, and $(1-\alpha^{s_b})[1-\pi+\pi q(1-F^{s_b})](r-c_b) = \mathbf{\Pi}^{s_b}$ with $\mathbf{\Pi}^{s_b}$ from (5) gives the distribution; $1-F^{s_b}(\underline{r}^b) = m$ gives $\underline{r}^b$. At $\bar{r}^g$, $F^{s_g} = 1$ and $F^{s_b} = 1-m$, so by (3), using $A^s + B^s = r - c^{avg,s}$, $\Pi^{s_b}(\bar{r}^g) = (1-\pi)(\bar{r}^g - c^{avg,s_b}) + \pi m B^{s_b}(\bar{r}^g)$. Setting this equal to $\mathbf{\Pi}^{s_b} = (1-\pi)(\bar{r}^g - c^{avg,s_b}) - (1-\pi)\phi^{s_b}$ gives (7), bounded by $m \in [0,1]$. On $(\bar{r}^g, \underline{r}^b)$ a bank earns $(1-\alpha^{s_b})[1-\pi+\pi q m](r-c_b)$ in its bad-signal group, which reaches $\mathbf{\Pi}^{s_b}$ only at $\underline{r}^b$.

Proposition 5. By Lemma 2, Φ_g and Φ_b are constant on a range quoted to both groups. In each configuration of Proposition 7 that range ends at $\bar{r}^g$, where $F^{s_g} = 1$ and $F^{s_b} = 1-m$ give $Y_g = (1-q)m$ and $Y_b = qm$, so the constants are $\Phi_g^\top$ and $\Phi_b^\top$. Inverting (2) gives $1 - Y_\theta = [(r-c_\theta) - \Phi_\theta^\top]/[\pi(r-c_\theta)]$, and inverting the two equations for Y_g, Y_b gives (8). Where only the good-signal group is quoted, $F^{s_b} = 0$ and (3) is linear in F^{s_g}, which gives (9); $F^{s_g} = 0$ gives $\underline{r}$.

Proposition 6. Differentiating (8), $\pi(2q-1)\,F^{s_g\prime} = q\Phi_g^\top/(r-c_g)^2 - (1-q)\Phi_b^\top/(r-c_b)^2$ and $\pi(2q-1)\,F^{s_b\prime} = q\Phi_b^\top/(r-c_b)^2 - (1-q)\Phi_g^\top/(r-c_g)^2$. So $F^{s_g\prime} \geq 0$ if and only if $[(r-c_b)/(r-c_g)]^2 \geq \tau^2$, that is $r \geq \rho_-$, because $(r-c_b)/(r-c_g)$ rises with r. $F^{s_b\prime} \geq 0$ if and only if $(\Phi_g^\top/\Phi_b^\top)[(r-c_b)/(r-c_g)]^2 \leq q/(1-q)$. The left side rises with r and at $\bar{r}^g$ equals $\frac{1-\pi+\pi(1-q)m}{1-\pi+\pi qm} \cdot \frac{\bar{r}^g - c_b}{\bar{r}^g - c_g} < 1 < \frac{q}{1-q}$, so this always holds. At $\bar{r}^g$, $F^{s_g} = 1$ by construction.

Proposition 7. By Proposition 1 the market opens with the good-signal group quoted alone, on (9). That phase ends when the quotes to the group run out, at ρ_g, or when quoting the bad-signal group becomes worthwhile. At the latter rate both indifference conditions hold with $F^{s_b} = 0$, so (9) and (8) agree there: it is ρ. If $\rho_g < \rho$, nobody would quote just above ρ_g, contradicting Lemma 4. Raising $\mathbf{\Pi}^{s_g}$ lowers (9) at every rate until the good-signal group's range ends where the bad-signal group's begins, with $F^{s_g} = 1$ and $F^{s_b} = 0$; then $1-\pi+\pi Y_g = L$ and $1-\pi+\pi Y_b = M$, and $\Pi^{s_b}(\rho) = \mathbf{\Pi}^{s_b}$, $\Pi^{s_g}(\rho) = \mathbf{\Pi}^{s_g}$ give configuration (i). Above ρ, (3) with $F^{s_g} = 1$ is linear in F^{s_b}, and $(r - c^{avg,s_b}) - \pi A^{s_b}(r) = \alpha^{s_b} L(r-c_g) + (1-\alpha^{s_b})M(r-c_b)$. If $\rho_g \geq \rho$, the two branches meet at ρ with $F^{s_g} < 1$ and $\mathbf{\Pi}^{s_g} = \mathbf{\Pi}^\circ$. If $\rho \geq \rho_-$, F^{s_g} rises on (8) from ρ on: configuration (ii). If $\rho < \rho_-$, sharing would need F^{s_g} to fall, so quotes to the good-signal group stop at $\bar{F}$ and only the bad-signal group is quoted, with (3) linear in F^{s_b}. Sharing resumes where (8) returns to $\bar{F}$; by Proposition 6, (8) falls until ρ_- and then rises to one, so it crosses $\bar{F} < 1$ once more, at $r_2 > \rho_-$. Above r_2, F^{s_g} rises, so there is no second pause: configuration (iii). If $m = 1$, the bad-signal group is not quoted in the pooling region, and (9) with $\mathbf{\Pi}^{s_g} = \mathbf{\Pi}^\circ$ reaches one at $\bar{r}^g$: configuration (iv).

Proposition 8. Since $\sigma_g\alpha^{s_g} = q_1\mu_g$, $\sigma_g(1-\alpha^{s_g}) = (1-q_1)\mu_b$, $\sigma_b\alpha^{s_b} = (1-q_1)\mu_g$ and $\sigma_b(1-\alpha^{s_b}) = q_1\mu_b$, quoting r_g to the good-signal group and r_b to the bad-signal group earns $q_1\mu_g\Phi_g(r_g)+(1-q_1)\mu_b\Phi_b(r_g)+(1-q_1)\mu_g\Phi_g(r_b)+q_1\mu_b\Phi_b(r_b)$, which is linear in q_1. Π^{dev} is the largest of these lines, so it is convex, and its one-sided slopes are the slopes of the lines through the best rates, (11). On the support of F^s, $\alpha\Phi_g + (1-\alpha)\Phi_b = \mathbf{\Pi}^s + (\alpha-\alpha^s)\Delta$, so for a share of good firms just above α^s the best rate in the support maximizes Δ, and just below it minimizes Δ. Just above q^* the share of good firms rises in the good-signal group and falls in the bad-signal group, which gives r_g^+ and r_b^+; just below, the reverse. On the good-signal group's range Δ is largest at $\underline{r}$, where it equals $c_b - c_g$. For monotonicity, write $X = \mu_g[\Phi_g(r_g) - \Phi_g(r_b)]$ and $Z = \mu_b[\Phi_b(r_g) - \Phi_b(r_b)]$. Optimality of r_g in the good-signal group and of r_b in the bad-signal group give $q_1X + (1-q_1)Z \geq 0$ and $(1-q_1)X + q_1Z \leq 0$. Subtracting, $(2q_1-1)(X-Z) \geq 0$, so the slope $X - Z$ is non-negative.

Proposition 9. W is the largest of functions linear in α, so it is convex. If $r^\sharp$ is best at both $\alpha^{s_b}(q^*)$ and $\alpha^{s_g}(q^*)$, it is best at every share in between. Less precision moves both shares inside that interval, so the deviator quotes $r^\sharp$ to both groups and, by (1), earns $\mu_g\Phi_g(r^\sharp)+\mu_b\Phi_b(r^\sharp)$. If no rate is best at both ends, W is not linear on the interval, so it lies strictly below its chord inside the interval. $\Pi^{dev}(q_1)$ is an average of W at two interior shares with mean μ_g, which is below the chord at μ_g, that is below $\Pi^{dev}(q^*)$. If $m < 1$, the meeting rate (configuration (i)) or any rate on a shared range (configurations (ii) and (iii)) is quoted to both groups. If $m = 1$, the good-signal group's range ends at $\bar{r}^g$ and the bad-signal group's range starts at $\underline{r}^b > \bar{r}^g$, so no rate is shared.

Proposition 10. Necessity: if q^* is best and C is strictly increasing, then for $q_1 < q^*$, $\Pi^{dev}(q^*) - \Pi^{dev}(q_1) \geq C(q^*) - C(q_1) > 0$. Sufficiency: take C piecewise linear through q^* with slope S_{down} below and S_{up} above. Both slopes are positive, and convexity of Π^{dev} gives $S_{up} \geq S_{down}$, so C is strictly increasing and convex. Convexity also makes the chord slope $[\Pi^{dev}(q^*) - \Pi^{dev}(q_1)]/(q^* - q_1)$ rise with q_1, so it is at least S_{down}, and $C(q^*) - C(q_1) \leq \Pi^{dev}(q^*) - \Pi^{dev}(q_1)$. The argument above q^* is the same.

Proposition 11. At $q^* = \frac{1}{2}$ there is no lower precision, and a steep enough cost deters any increase. For $q^* > \frac{1}{2}$, Proposition 9 gives a flat Π^{dev} below q^* when $m < 1$, which violates Proposition 10, and $\Pi^{dev}(q_1) < \Pi^{dev}(q^*)$ below q^* when $m = 1$, which satisfies it. By (7), $m = 1$ is $\phi^{s_b} \leq -\frac{\pi}{1-\pi}B^{s_b}(\bar{r}^g)$.

Proposition 12. At $q_1 = \frac{1}{2}$, $\sigma_g = \sigma_b = \frac{1}{2}$, both shares equal μ_g, and $d\alpha^{s_g}/dq_1 = -d\alpha^{s_b}/dq_1 = 4\mu_g\mu_b$. Differentiating (10), the terms in $W(\mu_g)$ cancel and $D^+(\frac{1}{2}) = 2\mu_g\mu_b[W'_+(\mu_g) - W'_-(\mu_g)]$. By the argument in Proposition 8, $W'_+(\mu_g)$ and $W'_-(\mu_g)$ are the largest and smallest premium over the rates quoted at $q = \frac{1}{2}$. With one distribution F, $\Delta = (1-\pi F)(c_b - c_g)$ in the pooling region and $\Delta = -\Phi_b$ in the bad-only region. The largest premium is $c_b - c_g$, at the lowest rate. Under full pooling the smallest is

$(1-\pi)(c_b - c_g)$, at $\bar{r}^g$. Under hybrid lending it is $-(1-\pi)(\bar{r}^b - c_b)$, anywhere in the bad-only range. Under bad-only lending every quoted rate has premium $-(1-\pi)(\bar{r}^b - c_b)$, so the difference is zero.

Perfect signal. At $q^* = 1$ in configuration (iv), apply the closed form for D^- in Proposition 8 with $q^* = 1$.

Appendix B. A numerical example

B.1 The economy

Appendices B.2 and B.3 use one economy: $\mu_g = 0.65$, $q = 0.75$, $c_g = 0.05$, $c_b = 0.30$, $\pi = 0.4$, $\bar{r}^g = 0.45$, $\bar{r}^b = 0.80$. The signal gives $\sigma_g = 0.575$, $\alpha^{s_g} = 0.848$ and $\alpha^{s_b} = 0.382$, and the banks disagree on 37.5% of firms.

B.2 Profit counted by the rival's group

Section 2.2 states that profit cannot be computed as an average chance of winning times an average margin. Take the 575 firms, out of 1,000 in this economy, in the bank's good-signal group, at $r = 0.40$, when the rival quotes its whole good-signal group below r and its whole bad-signal group above. The bank keeps a firm with probability 0.6 if the firm is in the rival's good-signal group and with probability 1 if it is in the rival's bad-signal group. The margin is 0.35 on a good firm and 0.10 on a bad one. Each cell is (firms) $\times$ (chance of keeping) $\times$ (margin):

	rival's good-signal group	rival's bad-signal group	total
good firm	$365.6 \times 0.6 \times 0.35 = 76.8$	$121.9 \times 1 \times 0.35 = 42.7$	119.4
bad firm	$21.9 \times 0.6 \times 0.10 = 1.3$	$65.6 \times 1 \times 0.10 = 6.6$	7.9
total	78.1	49.2	127.3

- *Adding by rows* is Lemma 1. Within a row the margin is fixed and only the chance of keeping is averaged: 0.7 for good firms and 0.9 for bad firms, which are $1-\pi+\pi Y_g$ and $1-\pi+\pi Y_b$. Profit per firm is $0.848 \times 0.7 \times 0.35 + 0.152 \times 0.9 \times 0.10 = 0.221$.
- *Adding by columns* weights the rival's group. The rival places the firm in its good-signal group with probability $387.5/575 = 0.674$, and then the average margin is 0.336. It places it in its bad-signal group with probability 0.326, and then the average margin is 0.2625. Profit per firm is $0.674 \times 0.6 \times 0.336 + 0.326 \times 1 \times 0.2625 = 0.221$.

B.3 Solving one equilibrium

The seven steps that solve the model, in the economy of B.1.

step	quantity	value
1	σ_g; α^{s_g}, α^{s_b}; c^{avg,s_g}, c^{avg,s_b}	0.575; 0.848, 0.382; 0.088, 0.204
2	ϕ^{s_b}; $\mathbf{\Pi}^{s_b} = (1-\alpha^{s_b})(1-\pi)(\bar{r}^b - c_b)$	-0.063; 0.185
3	$B^{s_b}(\bar{r}^g)$; m; $\underline{r}^b$	0.108; 0.881 (hybrid); 0.647
4	$\Phi_g^{\top}$, $\Phi_b^{\top}$; $\mathbf{\Pi}^{\circ}$	0.275, 0.130; 0.2531
5	ρ; ρ_g (with $L = 0.7$, $M = 0.9$)	0.4441; $0.4433 < \rho$: configuration (i)
6	meeting rate ρ; $\mathbf{\Pi}^{s_g}$	0.4437; 0.2534
7	(A): $\mathbf{\Pi}^{s_g} \geq (1-\alpha^{s_g})(1-\pi)(\bar{r}^b - c_b) = 0.046$	holds; $\phi^{s_g} = 0.286$

In this equilibrium:

- The good-signal group is quoted rates on $[0.341, 0.444]$.
- The bad-signal group is quoted rates on $[0.444, 0.45]$, where F^{s_b} reaches $1 - m = 0.119$, and on $[0.647, 0.80]$. Nobody quotes on $(0.45, 0.647)$.
- Expected profit per firm examined is $\mathbb{E}[\Pi(q)] = 0.224$.

B.4 Checking the equilibrium at every rate

This check uses the economy of Section 4.3 at $q = 0.80$, the equilibrium in which the good-signal group pauses.

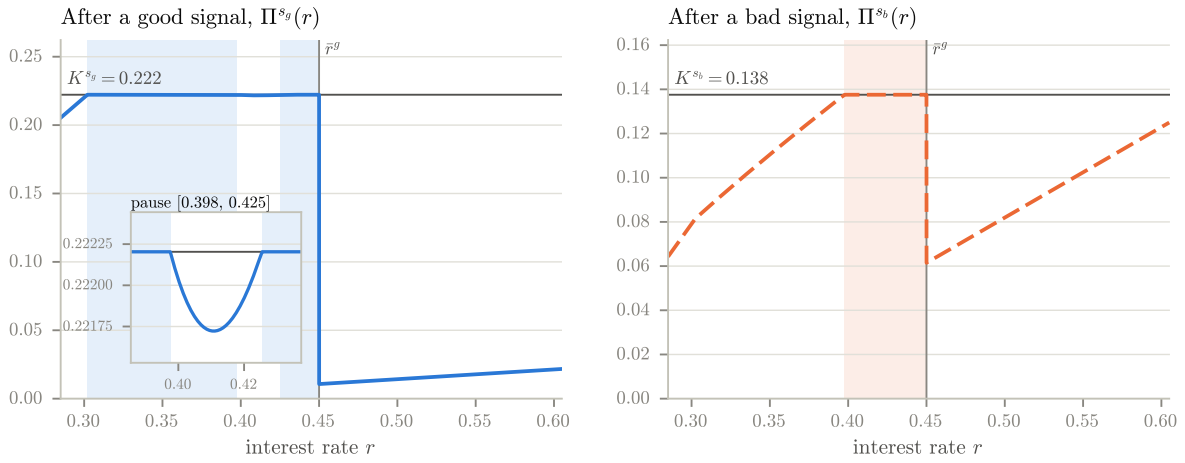


Figure 5: Profit in each signal group at every rate, against the equilibrium distributions at $q = 0.80$ (the pause). Shaded: rates quoted to that group. Horizontal line: equilibrium profit $\mathbf{\Pi}^s$. Vertical line: $\bar{r}^g$. Inset: the pause. Economy of Section 4.3.

The equal-profit condition holds: profit equals $\mathbf{\Pi}^s$ wherever the group is quoted and is lower everywhere else.

- **Good-signal group.** Below the lowest quote (0.302) profit rises with the rate. During the pause it dips below $\mathbf{\Pi}^{s_g}$ by 0.0005 (0.2%): too little to see at full scale,

but enough that no bank quotes its good-signal group there. In the bad-only region good firms reject, and profit falls to what bad firms alone pay.

- **Bad-signal group.** Profit rises until $\rho = 0.398$, stays at $\mathbf{\Pi}^{s_b}$ up to $\bar{r}^g$, drops at $\bar{r}^g$ as the good firms in the group reject, and climbs back only to 0.123 at $\bar{r}^b = 0.60$. That is why the bad-signal group is not quoted in the bad-only region at this q.

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